Experiment Proposal: Active Damping of Vacuum Fluctuations (Higgs Damping)

Principal Investigator: Peter Kyjovsky **Experiment Type:** Laboratory (Tabletop) / Low-Temperature Physics

1. Objective

To verify the feasibility of manipulating the "zero point" of the Higgs field using a coherent electromagnetic field within a superconducting cavity. The goal is to reduce the density of virtual particles (lattice vibrations) below the ground state level (Squeezed Vacuum State).

2. Hypothesis

If the Higgs field is a crystalline lattice with a resonant frequency, then applying an external field in precise anti-phase will cause:

1. A change in vacuum permittivity (\epsilon_0) inside the cavity.
2. A measurable shift in the cavity's resonant frequency (deviation from QED prediction).

3. Hardware Configuration

- **Environment:** Dilution Refrigerator cooled to 10-20 mK.
- **Resonator:** Niobium superconducting cavity with a high quality factor (Q > 10^{10}).
- **Driver:** Microwave or Terahertz radiation generator, tuned to harmonic frequencies of the Higgs Lattice.
- **Sensor:** SQUID magnetometer or optical interferometer (homodyne detection).

4. Measurement Procedure

Phase 1: Calibration (Baseline)

1. Cool the system to operating temperature.
2. Measure the thermal noise spectrum and fundamental vacuum noise (Shot noise) without excitation.

Phase 2: Frequency Scanning (Hunting)

1. Activate the driver and slowly sweep the frequency.
2. Search for "anomalous absorptions"—points where the cavity absorbs energy, but it does not convert to heat (energy "disappears" into the Higgs Lattice - tunneling).

Phase 3: Active Damping

1. Set frequency to the identified resonance.
2. Apply a signal with a phase shift of 180^{\circ} (anti-phase).
3. Monitor noise levels. If the theory holds, noise should drop **below the Standard Quantum Limit**.

5. Expected Result

A decrease in measured noise within the cavity would indicate that we have successfully "calmed" the local Higgs Lattice and created a microscopic region of stabilized vacuum (a tunnel seed).